		Hence, effective resistance = $\frac{1}{2}R$ = 0.50 Ω
23	**D**	Since the currents are in opposite directions, the magnetic fields between the wires are in the same direction, so they cannot cancel there. Hence the zero-field point must lie outside the two wires. It must be on the right-hand side of the right wire, because the right wire carries the smaller current 3.0 A, so the point must be closer to it for its field to match that of the 4.0 A wire. Let the distance of the point from left wire be y $\frac{\mu_0 I_x}{2\pi d} = \frac{\mu_0 I_y}{2\pi d}$ $\frac{4.0}{y} = \frac{3.0}{(y-d)}$ $y = 4d$
24	**B**	For charged particles moving perpendicular in a magnetic field, the magnetic force will provide centripetal force and their radius will be